

Intermediate Microeconomics

Week 3 · Day 3 — Repeated Games

Summer School

The Prisoner's Dilemma said cooperation is doomed. Repetition — and the shadow of the future — can rescue it.

Yesterday's puzzle

In the one-shot price war, cutting price is a dominant strategy, so the firms are trapped at (Low, Low) = (60, 60), forgoing the collusive (100, 100).

	High	Low
High	(100, 100)	(20, 140)
Low	(140, 20)	(60, 60)

But firms meet again and again

A supermarket sets prices *every week*; airlines compete *every day*. If I undercut you today, you can punish me tomorrow.

The question

Does the **fear of retaliation** let firms sustain the good outcome (100, 100)? It depends on whether the game ends at a *known date* or carries on indefinitely.

Learning goals — Day 3

- ① Set up a **repeated game**: a stage game played many times, with strategies that condition on *history*.
- ② Show that a **finitely** repeated Prisoner's Dilemma *unravels* to defection by backward induction.
- ③ Discount the future with a factor δ and value an infinite stream of payoffs.
- ④ Use a **grim-trigger** strategy to sustain cooperation in an **infinitely** repeated game.
- ⑤ Derive the **critical discount factor** δ^* and interpret it.
- ⑥ Apply it: sustaining collusion in the Day-1 duopoly, and the idea of the **Folk Theorem**.

What changes when a game is repeated

Ingredients of a repeated game

- A **stage game** (e.g. the price-war PD), played in periods $t = 1, 2, \dots$
- Players *observe* the past — so a strategy can be **history-dependent**: “cooperate, *unless* you ever cheated me.”
- Payoffs are summed *across periods*, with later payoffs discounted.

The key new freedom

Today's choice can be made *conditional* on the rival's past behaviour. Punishment becomes possible — and with it, the possibility of *rewarding* cooperation.

Two worlds, opposite answers

A **finite** horizon (a known last period T) versus an **infinite** horizon (it always might continue).

Finite repetition unravels — the logic

Suppose the price war is played exactly $T = 10$ times, and both firms know it. **Start at the end (period 10)**

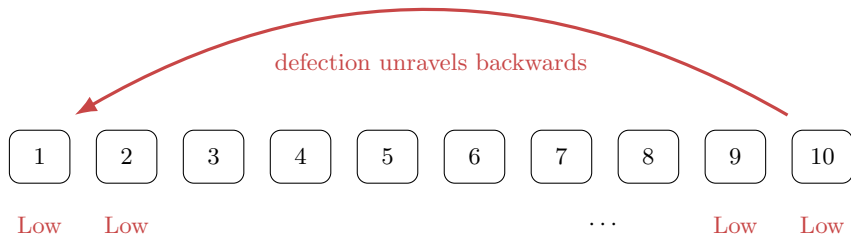
Period 10 is the *last* game — no future, so no possible punishment. It is just the one-shot PD: both play **Low**. **Now period 9**

Both know period 10 will be (Low, Low) *regardless* of what happens now. So period 9 also has no future consequence $\Rightarrow$ both play **Low**.

...and so on back to period 1

The same logic cascades backwards through every period. *Every* period is Low.

Finite repetition unravels — the picture



The known last period is certain to be non-cooperative — and that certainty pins down every earlier period.

The lesson of finite repetition

Result

If the stage game has a *unique* Nash equilibrium, then any **finite** repetition has that same equilibrium in *every* period. Cooperation cannot be sustained by backward induction.

- The problem is the *known* endpoint: the last period is unavoidably non-cooperative, and that certainty poisons every period before it.
- Intuition: you never cooperate in the final round, so there is no reward for cooperating in the second-to-last, and so on.

So what rescues cooperation?

Remove the known endpoint. If the game *never has a definite last period*, backward induction has nowhere to start — and the future can discipline the present.

Valuing the future: the discount factor

When the game may continue forever, we value a stream of payoffs $\pi_1, \pi_2, \dots$ with a **discount factor** $\delta \in (0, 1)$:

$$V = \pi_1 + \delta \pi_2 + \delta^2 \pi_3 + \dots = \sum_{t=1}^{\infty} \delta^{t-1} \pi_t.$$

Two readings of δ

- *Patience / interest*: $\delta = \frac{1}{1+r}$ — a dollar next period is worth δ today.
- *Continuation*: δ can also be the *probability the game continues* to the next period.

A constant stream is a geometric series

Getting π *every* period is worth $\pi + \delta\pi + \delta^2\pi + \dots = \frac{\pi}{1-\delta}$. This one formula does all the work below.

The grim-trigger strategy

Grim trigger

“Play **High** (cooperate) as long as everyone always has. The *first* time anyone plays Low, switch to **Low** forever after.”

The threat: one defection triggers a permanent price war — reversion to the stage-game outcome (60,60) for the rest of time. A firm compares two options:

Option 1 — cooperate forever

Earn 100 every period: $V_{\text{coop}} = \frac{100}{1 - \delta}$.

Option 2 — defect today

Grab 140 now, then get punished with 60 forever: $V_{\text{dev}} = 140 + \frac{\delta \cdot 60}{1 - \delta}$.

The critical discount factor — the algebra

Cooperation is sustainable when it is *not worth* deviating: $V_{\text{coop}} \geq V_{\text{dev}}$,

$$\frac{100}{1-\delta} \geq 140 + \frac{60\delta}{1-\delta}.$$

Multiply through by $(1-\delta) > 0$

$$100 \geq 140(1-\delta) + 60\delta = 140 - 80\delta.$$

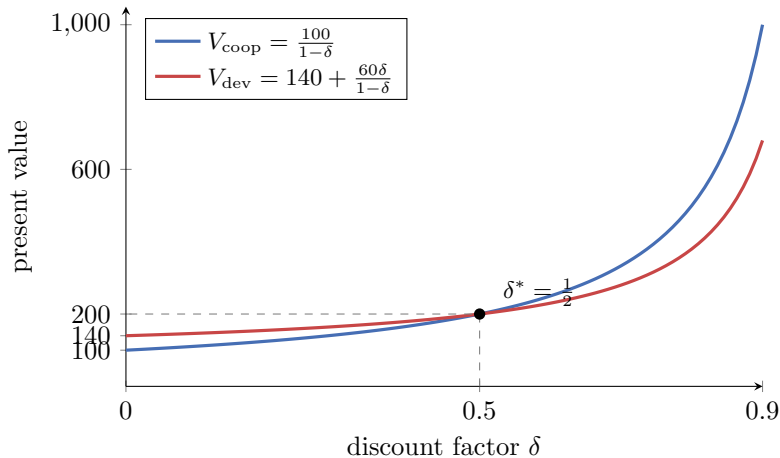
Solve for δ

$$80\delta \geq 40 \implies \boxed{\delta \geq \frac{1}{2}}.$$

If firms value the future enough ($\delta \geq \frac{1}{2}$), the fear of a permanent price war sustains the collusive (100, 100) *every* period.

Numeric check: at $\delta = \frac{1}{2}$: $V_{\text{coop}} = \frac{100}{0.5} = 200$ and $V_{\text{dev}} = 140 + \frac{60(0.5)}{0.5} = 200$ — exactly indifferent. ✓

The critical discount factor — the picture



For $\delta > \frac{1}{2}$ the cooperation curve lies above the deviation curve — cheating does not pay.

Reading the threshold

- $\delta \geq \frac{1}{2}$: patient firms, or a game very likely to continue $\Rightarrow$ collusion holds.
- $\delta < \frac{1}{2}$: impatient firms, or a game likely to end soon $\Rightarrow$ the temptation to grab 140 today wins, and price competition returns.

What makes collusion *harder* (raises the critical δ)?

- a bigger one-time gain from cheating (larger 140);
- a milder punishment (higher post-defection profit);
- fewer, less frequent interactions, or a nearer horizon.

This is why antitrust authorities watch *concentrated, stable, transparent* markets: there the future looms large and tacit collusion is easiest to sustain.

Application: colluding in the Day-1 duopoly

Recall the Cournot market ($P = 120 - Q$, $c = 30$). The firms could instead *split the monopoly output*: each makes 22.5, price 75, earning $\frac{2025}{2} = 1012.5$ each — better than the Cournot 900.

Temptation to cheat

If the rival sticks to 22.5, the best deviation is $q = \frac{90-22.5}{2} = 33.75$, giving price 63.75 and

$$\pi_{\text{dev}} = (63.75 - 30) \cdot 33.75 \approx 1139.$$

Punishment: revert to Cournot forever, 900 each.

Sustainability condition

$$\frac{1012.5}{1 - \delta} \geq 1139 + \frac{900 \delta}{1 - \delta} \implies 1012.5 \geq 1139 - 239\delta \implies \delta \geq \frac{126.5}{239} \approx 0.53.$$

If the firms are patient enough ($\delta \gtrsim 0.53$), the *repeated* game supports the monopoly outcome — the competition we found on Day 1 was a feature of the *one-shot* game.

The Folk Theorem — an informal statement

The idea

In an infinitely repeated game with patient enough players (δ close to 1), *almost any* outcome that gives each player more than their punishment (“safety”) payoff can be sustained as an equilibrium — including full cooperation.

Good news

Cooperation, collusion, and rich self-enforcing agreements are all *possible* — with no external enforcement.

Cautionary news

So are many *other* outcomes. Repetition creates a *multiplicity* of equilibria; theory alone no longer pins down a single prediction.

The single force behind all of it: a credible *future* reward or punishment, possible only because the game has no known end.

Finite vs. infinite — side by side

	Finitely repeated	Infinitely repeated
Known last period?	Yes	No
Solved by	Backward induction	Compare PV streams
Can cooperation survive?	No — unravels	Yes, if $\delta \geq \delta^*$
Driving force	Certain endpoint	Shadow of the future
Our example	Low every period	High if $\delta \geq \frac{1}{2}$

The knife-edge is the *endpoint*, not the number of rounds: it is the *certainty* of a last game that destroys cooperation, and its *absence* that restores it.

Day 3 checkpoint

You can now

- set up a repeated game with history-dependent strategies and discounting;
- show a finitely repeated PD unravels to defection;
- value an infinite payoff stream with $\frac{\pi}{1-\delta}$;
- test cooperation with a grim trigger and solve for the critical δ^* ;
- apply it to sustain collusion in a duopoly, and state the Folk Theorem.

Tomorrow — a different kind of failure

So far markets failed through *market power*. Day 4: markets that fail even under competition — **externalities**, **public goods**, and the **tragedy of the commons**.

In-class exercises — Day 3

Time for the in-class exercises.

- **Exercise 1** — finite repetition: show a 3-round PD unravels, and explain the role of the endpoint.
- **Exercise 2** — infinite repetition: derive the critical discount factor for a given PD, and see how it shifts when the temptation or punishment changes.

Work in pairs; be ready to state your δ^* and what would make cooperation *harder*.